THE CLOSURE OF THE GRAPH OF A RATIONAL FUNCTION IN A PROJECTIVE LINE BUNDLE: PROPERNESS, THE TWO PULLBACK DIVISORS AND THE DIVISOR OF F

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Composition of proper morphisms. Morphisms, label morphisms-lemma-composition-proper, invoked at proof line 2746. Makes the projection p : Y -> X proper as the composite of the closed immersion Y -> P^1_X with the proper projection P^1_X -> X; this is assertion (1).
- Pullback of an effective Cartier divisor along a morphism. Divisors, labels divisors-definition-pullback-effective-Cartier-divisor (statement, proof line 2716) and divisors-lemma-pullback-effective-Cartier-defined (proof line 2756). Defines Y_0 and Y_infty as pullbacks of the two standard sections and supplies the local criterion that makes them defined because q(eta') avoids D_0 and D_infty; assertions (3) and the dimension count following it are built on this.
- Dimension of the support of a divisor and compatibility of pushforward with divisors of rational functions. Same chapter chow.tex, labels lemma-divisor-delta-dimension (proof line 2759), lemma-cycle-push-sheaf (proof line 2790) and lemma-proper-pushforward-alteration (proof line 2794). Gives the delta-dimension of Y_0 and Y_infty, identifies p_*[Y_0]_{n-1} with the class of the image of Y_0, and supplies the compatibility of p_* with div(f) that yields assertion (5).

Source reference. Original lemma 01WV: sources/stacks-chow/tags.txt, source lines 2496–2496. Navigation pointer for \ref{divisors-definition-pullback-effective-Cartier-divisor} used inside the counted span (line 2716); the argument that consumes it is retained in the span.

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Source reference. Original lemma 02OO: sources/stacks-chow/tags.txt, source lines 3523–3523. Navigation pointer for \ref{divisors-lemma-pullback-effective-Cartier-defined} used inside the counted span (line 2756); the argument that consumes it is retained in the span.

Source reference. Original lemma 02R6: sources/stacks-chow/pinned-original-source.tex, source lines 1992–1992. Navigation pointer for \ref{lemma-cycle-push-sheaf} used inside the counted span (line 2790); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RK: sources/stacks-chow/pinned-original-source.tex, source lines 2489–2489. Navigation pointer for \ref{lemma-divisor-delta-dimension} used inside the counted span (line 2759); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RT: sources/stacks-chow/pinned-original-source.tex, source lines 2645–2645. Navigation pointer for \ref{lemma-proper-pushforward-alteration} used inside the counted span (line 2794); the argument that consumes it is retained in the span.

Source reference. Original lemma 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Navigation pointer for \ref{morphisms-lemma-composition-proper} used inside the counted span (line 2746); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2702); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Assume X is integral and $n = \dim_\delta(X)$. Let $f \in R(X)^*$. Let $U \subset X$ be a nonempty open such that f corresponds to a section $f \in \Gamma(U, \mathcal{O}_X^*)$. Let $Y \subset X \times_S \mathbf{P}^1_S$ be the closure of the graph of $f : U \to \mathbf{P}^1_S$. Then*

1. *the projection morphism $p : Y \to X$ is proper,*
2. *$p|_{p^{-1}(U)} : p^{-1}(U) \to U$ is an isomorphism,*
3. *the pullbacks $Y_0 = q^{-1}D_0$ and $Y_\infty = q^{-1}D_\infty$ via the morphism $q : Y \to \mathbf{P}^1_S$ are defined (Divisors, Definition 01WV),*
4. *we have*
$$div_Y(f) = [Y_0]_{n-1} - [Y_\infty]_{n-1}$$
5. *we have*
$$div_X(f) = p_* div_Y(f)$$
6. *if we view Y_0 and Y_∞ as closed subschemes of X via the morphism p then we have*
$$div_X(f) = [Y_0]_{n-1} - [Y_\infty]_{n-1}$$

Proof. Since X is integral, we see that U is integral. Hence Y is integral, and $(1, f)(U) \subset Y$ is an open dense subscheme. Also, note that the closed subscheme $Y \subset X \times_S \mathbf{P}^1_S$ does not depend on the choice of the open U, since after all it is the closure of the one point set $\{\eta'\} = \{(1, f)(\eta)\}$ where $\eta \in X$ is the generic point. Having said this let us prove the assertions of the lemma.

For (1) note that p is the composition of the closed immersion $Y \to X \times_S \mathbf{P}^1_S = \mathbf{P}^1_X$ with the proper morphism $\mathbf{P}^1_X \to X$. As a composition of proper morphisms is proper (Morphisms, Lemma 01W3) we conclude.

It is clear that $Y \cap U \times_S \mathbf{P}^1_S = (1, f)(U)$. Thus (2) follows. It also follows that $\dim_\delta(Y) = n$.

Note that $q(\eta') = f(\eta)$ is not contained in D_0 or D_∞ since $f \in R(X)^*$. Hence (3) by Divisors, Lemma 02OO. We obtain $\dim_\delta(Y_0) = n-1$ and $\dim_\delta(Y_\infty) = n-1$ from Lemma 02RK.

Consider the effective Cartier divisor Y_0. At every point $\xi \in Y_0$ we have $f \in \mathcal{O}_{Y,\xi}$ and the local equation for Y_0 is given by f. In particular, if $\delta(\xi) = n-1$ so ξ is the generic point of an integral closed subscheme Z of δ-dimension $n-1$, then we see that the coefficient of $[Z]$ in $\mathrm{div}_Y(f)$ is

$$\mathrm{ord}_Z(f) = \mathrm{length}_{\mathcal{O}_{Y,\xi}}(\mathcal{O}_{Y,\xi}/f\mathcal{O}_{Y,\xi}) = \mathrm{length}_{\mathcal{O}_{Y,\xi}}(\mathcal{O}_{Y_0,\xi})$$

which is the coefficient of $[Z]$ in $[Y_0]_{n-1}$. A similar argument using the rational function $1/f$ shows that $-[Y_\infty]$ agrees with the terms with negative coefficients in the expression for $\mathrm{div}_Y(f)$. Hence (4) follows.

Note that $D_0 \to S$ is an isomorphism. Hence we see that $X \times_S D_0 \to X$ is an isomorphism as well. Clearly we have $Y_0 = Y \cap X \times_S D_0$ (scheme theoretic intersection) inside $X \times_S \mathbf{P}^1_S$. Hence it is really the case that $Y_0 \to X$ is a closed immersion. It follows that

$$p_*\mathcal{O}_{Y_0} = \mathcal{O}_{Y_0'}$$

where $Y_0' \subset X$ is the image of $Y_0 \to X$. By Lemma 02R6 we have $p_*[Y_0]_{n-1} = [Y_0']_{n-1}$. The same is true for D_∞ and Y_∞. Hence (6) is a consequence of (5). Finally, (5) follows immediately from Lemma 02RT. □